Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Multifunctional dual-release nanodiamond 3D-Ti-implants with programmable biological activity			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (x), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Medical/ Bio	Bio/ Biomaterials	Surface Treatment	Surface Property Modification Technologies
Concept of Proposed Technology	The core concept of the proposed technology is an innovative approach to reconstructing craniofacial defects by creating titanium (Ti)-based implants with programmable surface and biological properties. The implants will be engineered using (i) additive manufacturing 3D-printing technology and (ii) customized surface treatment method by using electrochemical surface modification to achieve the programmable biological activity, enhanced biocompatibility and osteointegration. Electrochemical surface modification will allow anchoring and immobilization of different nanoparticles, biomolecules and drugs, e.g. integration of chemically modified detonation nanodiamonds (NDs) and hydroxyapatite nanoparticles (HAP) into the titanium oxide layer. The integration of advanced biological validation and *in vivo* assessment will ensure the clinical reliability of these programmable surfaces. The selective application of different NDs functionalizations on different surface areas of the implant proposed in this project will enable the controlled interaction of the implant prototypes with hard and soft tissues. The development of such implants could significantly improve outcomes and rehabilitation prognoses in patients with complex maxillofacial defects.			

(max 10 pages)

	Craniofacial therapy • Extensive local therapeutic administration in case of craniofacial defects and ailments • Nano-engineered Ti-implants: bone therapy, antibacterial therapy, cancers and tumors, soft-tissue integration • Craniofacial therapy of periodontal diseases, trauma, tumours in the craniomaxillofacial region (resection surgery of tumors) or birth defects (microtia, anophthalmia) • Biosafety and biocompatibility • Osteoinductive or osteoconductive nature • Mechanical stability • Ease of fabrication and sterilization • Orchestrate both hard- and soft-tissue integration
Rationale for Joint R&D Project	Reconstructing maxillofacial defects is quite challenging due to the region's complex anatomy, and cosmetic and functional effects on patients. With the help of developing technologies, patient-specific implants (PSIs) using virtual surgical planning based on a computer aided designing (CAD)/Computer aided manufacturing (CAM) platform is an evolving treatment option. PSIs can be used in patients with maxillofacial defects and reconstruction. Skeletal and soft tissue structures are well organized in a complex three-dimensional (3D) architecture. Simulating this 3D-complexity and its multicellular interactions is one of the greatest challenges, especially for the functional reconstruction of large defects. Implants with programmable biological activity (osteointegration, antimicrobial and antioxidant properties, accelerated healing) – adapted to the unique structure of the facial skeleton – may enable a shift from substitutional to regenerative reconstruction. To address these constraints, there is a growing need for biomaterials like metal-based implants for improving damaged tissues' structural and functional integrity in biomedical applications. For craniofacial correction, ideally, the implants must meet the following requirements: • Biosafety and biocompatibility • Osteoinductive or osteoconductive nature • Mechanical stability • Ease of fabrication and sterilization • Orchestrate both hard- and soft-tissue integration. The proposed Korean-German project aims to develop an innovative approach to reconstructing craniofacial defects by creating titanium (Ti)-based implants with programmable surface and biological properties. The implants will be engineered using (i) additive manufacturing 3D-printing technology and (ii) surface biofunctionalization techniques to achieve the programmable biological activity, enhanced biocompatibility and osteointegration (Fig. 1), (iii) Physical-chemical, *in vitro* and *in vivo* biological assessment to obtain implant properties. The proposed joint Korean-German collaboration project is designed around the principle of complementary excellence, where each partner contributes a distinct core competency that is essential to de-risk integration and accelerate technology readiness. The proposed technology is inherently interdisciplinary and cannot be effectively realized through a single institutional capability. • Korea team contribution: Advanced 3D fabrication and multi-stage biological validation of patient-specific implants - Patient-Specific 3D Modeling & Virtual Planning: Leveraging CAD/CAM platforms for precise virtual surgical planning and designing geometrically complex implants tailored to individual craniofacial defects. - High-Precision Additive Manufacturing: Utilizing SLS (Selective Laser Sintering) and SLM technologies to fabricate titanium alloy implants with optimized porosity and mechanical integrity. - Advanced Biological & Pre-clinical Validation: Executing comprehensive in-vitro assays and in-vivo animal studies to evaluate bone regeneration, cell proliferation, and soft-tissue integration. - Quality Control & Physicochemical Analysis: Implementing rigorous testing protocols, including CT-based dimensional control and mechanical stability assessments, to ensure the reliability of 3D-printed prototypes. - Regulatory Compliance & Market Strategy: Leading the MFDS (KFDA) medical device certification

process and establishing a strategic roadmap for domestic and global commercialization within the medical device industry.

- German team contribution: Customized surface treatment method by using electrochemical surface modification
  - Electrochemical surface modification treatment: Utilizes electrochemical method of anodic polarization for immobilization and incorporation of DND and HAP in titanium oxide layers.
  - Biochemical functionalization of NDs: Different surface functionalization of NDs e.g using phosphate-based anchoring groups and adenosine triphosphate lead to higher affinity of NDs to titanium surfaces; alkyl chains (n-propylamine, n-butylamine etc.) increase the hydrophobicity and polyethylenglycole (PEG) und alkandiols (amino-propandiol etc.) - the hydrophilicity of NDs. The binding of Bone Morphogenetic Proteins (BMP-2) and antimicrobilal compounds such as Ag or Cu ions and antibioitics to NDs can ensure the antimicrobial properties.
  - Manufacturing of HAP: chemical precipitation method for the formation of mesoporous HAP nanoparticles, with the intent of changing and controlling the morphology and chemical properties of nanoparticles produced by the opposite ion core-shell method. The biological properties will be enhanced by the addition of different elements like magnesium and strontium.
  - Physical-chemical characterization (SEM, DLS, XRD, XPS, Raman, FTIR, OCT, speckle imaging and optical spectroscopy methods)
  - Cell-based in vitro testing
  - Bio- and Immune-compatibility studies

The proposed Korean–German joint R&D project is necessary to realize a clinically reliable nanodiamond 3D-Ti-implants with programmable biological activity, biocompatibility and required mechanical and osseoinductive properties. The Korean team brings validated pre-research capabilities in high-fidelity patient-specific 3D modeling and virtual surgical planning based on advanced CAD/CAM workflows. This includes the optimization of metal additive manufacturing processes—specifically SLS and SLM—to achieve precise trabecular structures for bone ingrowth. Furthermore, the team maintains established, high-standard protocols for evaluating biocompatibility, osteoinductivity, and soft-tissue integration using advanced bio-imaging, micro-CT analysis, and complex histological assessments in large animal models.

The German team contributes established expertise in electrochemical surface modification, nanofunctionalization, fabrication of HAP, physical and chemical surface characterization, cell-based in vitro testing, bio- and immune-compatibility studies . Since the 3D implant design and manufacturing as well as surface modification and characterization are strongly independent iterative co-design and verification through a bilateral collaboration is essential to reduce integration risk, improve reproducibility, and accelerate technology readiness toward clinical and field deployment.

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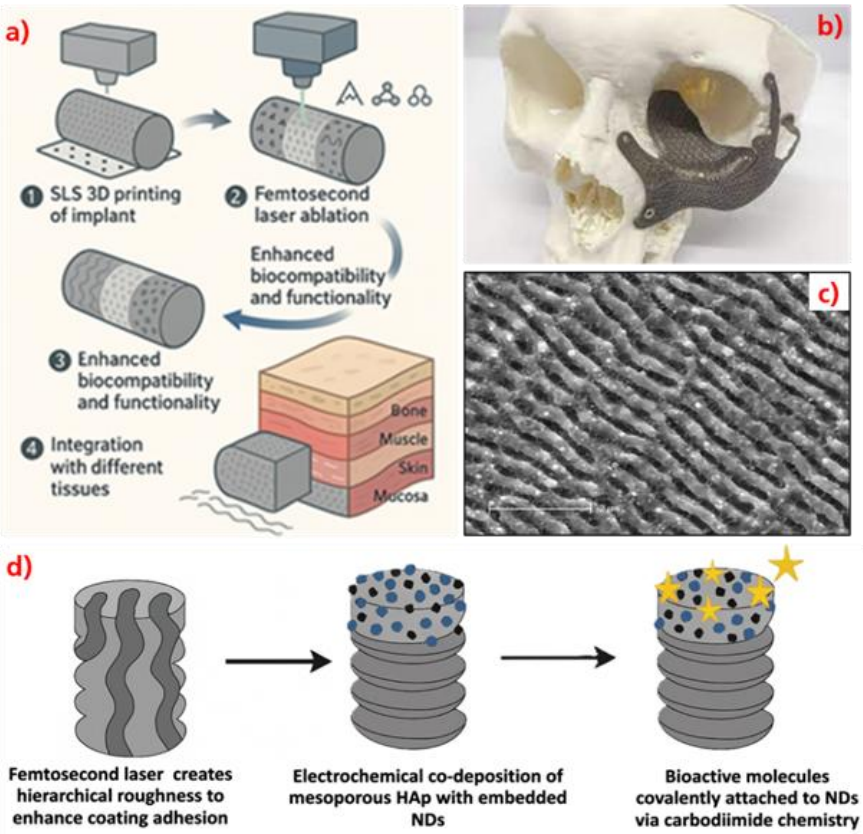

Fig. 1. Concept of multifunctional dual-release nanodiamond 3D-Ti-implants with programmable biological activity: a) general concept of implant manufacturing; b) individualized 3D design; c) surface treatment; d) individualized biochemical surface functionalization.

The developed implant prototypes will be tested *in vitro* and *in vivo* focusing on investigation of the cell adhesion and cell proliferation, antibacterial properties and reaction of the surrounding tissues after the implantation experiments *in vivo*. The main technological challenge of the project is to create an individualized implant surface having simultaneously different specific biological properties: implant surface dedicated to bone integration should promote osseointegration, while the surface intended for contact with soft tissues should support cell differentiation towards fibroblasts. Antibacterial activity of the fabricated surfaces is expected by selective surface engineering and smart antimicrobial biofunctionalization strategies.

Key Words

1. Reconstructing craniofacial defects
2. (Ti)-based implants with programmable surface and biological properties
3. Patient-specific 3D printing
4. Electrochemical surface modification treatment
5. *In vitro / in vivo* biological assessment and validation

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

Development of multifunctional dual-release nanodiamond 3D-Ti-implants with programmable biological activity where improved antioxidant, antibacterial, healing, and osteogenic properties will be achieved by selective incorporation of functionalized nanodiamonds and HAP nanoparticles. Different surface functionalization of NDs e.g using phosphate-based anchoring groups and adenosine triphosphate lead to higher affinity of NDs to titanium surfaces; alkyl chains (n-propylamine, n-butylamine etc.) increase the hydrophobicity and polyethylenglycole (PEG) und alkandiols (amino-propandiol etc.) - the hydrophilicity of NDs. The binding of Bone Morphogenetic Proteins (BMP-2) and antimicrobilal compounds such as Ag or Cu ions and antibioitics to NDs can ensure the antimicrobial properties. The selective application of different NDs functionalizations on different surface areas of the implant proposed in this project will enable the controlled interaction of the patient-specific Ti-based implant prototypes with hard and soft tissues. Furthermore, the project includes the optimization of additive manufacturing (SLS) processes for precise implant fabrication and comprehensive *in vitro* and *in vivo* biological validation using animal models to ensure clinical reliability and safety.

R&D Target Specifications

The following target performance specifications will be achieved to ensure the clinical reliability and functional superiority of the multifunctional 3D-printed titanium implants:

No.	Key Performance Indicator (KPI)	Target Specification	Measurement Method
1	Dimensional Accuracy	Deviation < ±100 µm from CAD	CT scan and 3D comparison analysis
2	Coating Adhesion Strength	> 20 MPa (at Ti/Coating interface)	Standard pull-off test (ASTM D4541)
3	Antimicrobial Efficacy	> 90% reduction	Bacterial colony counting (S. aureus, P. aeruginosa)
4	In vitro Cell Viability	> 90% (compared to bare Ti)	MTT/XTT assay and fluorescence imaging
5	Bone-to-Implant Contact (BIC)	> 50% increase in osseointegration	Histomorphometry in animal models
6	Surface Hydrophilicity Control	Contact angle: 10° ~ 110° (Programmable)	Water contact angle measurement

R&D Plan

Technical challenges of the project related to specific objectives:

- Fabrication of 3D sample prototypes using selective laser sintering (SLS). Titanium and its alloys have general applications as medical implant materials due to their superior mechanical properties and excellent biocompatibility. Novel 3D-printing of medical products has high speed, precision and good mechanical performance. Selective laser melting, one of the metal additive manufacturing processes, usually prepares single-component samples using a powder bed distribution system. The 3D-printed implants usually need to be processed further to achieve the desired surface roughness, density, hardness, strength, micro-structure, etc., to meet the specific functional requirements of maxillofacial and orbital cases. The proposed German-Korean collaboration addresses a challenge to develop tailored implants and optimise 3D-printing parameters to obtain surfaces promoting cell adhesion and differentiation. The primary objective is to fabricate geometrically complex sample prototypes (as predecessors for future patient-specific implants) with high precision and tailored mechanical and biological properties. The particular working steps include: design of implant geometry, material (powder) selection and preparation (powder morphology and particle size distribution analysis will be investigated by Scanning Electron Microscopy (SEM)), finding the right SLS process parameter, fabrication of 3D-sample prototypes. Dimensional control of the printed implant will be carried out by means of computed tomography (CT) and compared to the suggested 3D-models.

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- **Customized surface treatment method by using electrochemical surface modification**

Electrochemical surface modification treatment is a simple and versatile surface modification process for valve metals that can add superior osseointegration and bioactive properties to titanium and its alloys. This method is applicable in processing specimens of complicated geometry as the whole specimen is immersed in solution during the process. Electrochemical surface modification will allow anchoring and immobilization of different nanoparticles, biomolecules and drugs, e.g. integration of chemically modified detonation nanodiamonds (NDs) and hydroxyapatite nanoparticles (HAP) into the titanium oxide layer. The technical challenge of this task consist in the adjustment of electrochemical process to the treatment of Ti-based 3D-samples by preparing a suitable electrolyte solution and selection of optimal electrical parameters of electrochemical modification (current/voltage characteristics, chemical composition of electrolyte bath, optimal concentration of additives, temperature), which would enable to achieve sample surfaces with desired morphological, mechanical and biological properties including the incorporation of functionalized NDs and HAP.

- **Biochemical functionalization of NDs and manufacturing HAP nanocomposites**

NDs are a promising carbon-derived nanoscale material, which has been investigated since some decades due to its outstanding properties, including in the field of bone tissue engineering. Different surface functionalization of NDs e.g using phosphate-based anchoring groups and adenosine triphosphate lead to higher affinity of NDs to titanium surfaces; alkyl chains (n-propylamine, n-butylamine etc.) increase the hydrophobicity and polyethylenglycole (PEG) und alkandiols (amino-propandiol etc.) - the hydrophilicity of NDs. The binding of Bone Morphogenetic Proteins (BMP-2) and antimicrobilal compounds such as Ag or Cu ions and antibioitics to NDs can ensure the antimicrobial properties. The selective application of different NDs functionalizations on different surface areas of the implant proposed in this project will enable the controlled interaction of the implant prototypes with hard and soft tissues. In this way biocompatibility, cell viability, cell growth dynamics and antimicrobial properties of the sample surfaces will be provided in a controlled way. To further enhance the osteogenesis and biomechanical response HAP nanoparticles doped with magnesium and strontium ions will be synthetized and incorporated into the titanium oxide layer during electrochemical surface modification. The challenge of the biochemical functionalization of NDs and HAP consists in developing reproducible and effective functionalization schemes compartible with the following process of PEO treatment and applied sterilization tools. The functionality of the NDs and HAP should not be deteriorated by PEO.

- **Physical-chemical, *in vitro* and *in vivo* biological assessment of the sample properties**

The physical-chemical properties of the fabricated samples will be studied using SEM, XRD, XPS, Raman, FTIR, OCT, speckle imaging, and optical spectroscopy methods. In vitro tests will be done to assess seeding of osteoblasts, fibroblasts, and mesenchymal stem cells onto treated implant surfaces to investigate cell metabolic activity (MTT/XTT), cell adhesion (OCT), cell proliferation (speckle imaging/OCT), osteogenic markers (e.g., ALP, osteopontin), and calcium deposition (ARS/Raman/FTIR, fluorescence). Antibacterial efficacy screening will be performed using S. aureus and P. aeruginosa cultures. Bioactivity of the surfaces will be studied by in vitro apatite-forming ability assay. In vivo experiments will be conducted using appropriate animal models (e.g., rodents or rabbits) to replicate anatomical and physiological conditions of human maxillofacial regions. Functionalized implants will be surgically inserted, and biological responses—including bone-to-implant contact (BIC) and new bone formation—will be monitored over time by means of high-resolution bio-imaging and advanced histological analysis. Furthermore, ultrasound elastography for measuring the stiffness of tissues around the implant and digital volume tomography for assessing the biomechanical stability of implants will be applied to ensure the clinical reliability of the developed prototypes.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Contributions of Fraunhofer:** • Customized surface treatment method by using electrochemical surface modification - *Electrochemical surface modification treatment:* Utilizes electrochemical method of anodic polarization for immobilization and incorporation of DND and HAP in titanium oxide layers. - *Biochemical functionalization of NDs:* Different surface functionalization of NDs e.g using phosphate-based anchoring groups and adenosine triphosphate lead to higher affinity of NDs to titanium surfaces; alkyl chains (n-propylamine, n-butylamine etc.) increase the hydrophobicity and polyethylenglycole (PEG) und alkandiols (amino-propandiol etc.) - the hydrophilicity of NDs. The binding of Bone Morphogenetic Proteins (BMP-2) and antimicrobilal compounds such as Ag or Cu ions and antibioitics to NDs can ensure the antimicrobial properties. - *Manufacturing of HAP*: chemical precipitation method for the formation of mesoporous HAP nanoparticles, with the intent of changing and controlling the morphology and chemical properties of nanoparticles produced by the opposite ion core-shell method. The biological properties will be enhanced by the addition of different elements like magnesium and strontium. - Physical-chemical characterization (SEM, DLS, XRD, XPS, Raman, FTIR, OCT, speckle imaging and optical spectroscopy methods) - Cell-based *in vitro* testing - Bio- and Immune-compatibility studies.
Expected R&R of Korean SME Partner	The **Korean SME partner, in collaboration with a specialized research university**, will lead the localization, manufacturing, and commercialization phases of the project. Their primary responsibilities are divided into technical development and strategic validation as follows: • *Additive Manufacturing & Process Optimization:* Responsible for the high-precision fabrication of patient-specific titanium implants using Selective Laser Sintering (SLS) or Selective Laser Melting (SLM) technologies. • *Customized 3D Modeling:* Establishing a virtual surgical planning workflow based on CAD/CAM platforms to design complex geometries tailored to individual craniofacial defects • *Quality Control & Mechanical Testing:* Implementing rigorous dimensional accuracy checks using CT scanning and evaluating the mechanical integrity of 3D-printed prototypes to ensure clinical reliability. • Biological & Pre-clinical Validation: Leading the comprehensive in-vitro assays and in-vivo animal model studies to verify bone-to-implant contact (BIC), cell proliferation, and antibacterial efficacy of the functionalized surfaces. • *Regulatory Compliance & Market Access:* Managing the MFDS (KFDA) medical device certification process and developing commercialization strategies for both domestic and global medical device markets • *Resource Commitment:* Providing the necessary manufacturing infrastructure, including 3D printing facilities and human resources for technical integration and bilateral collaboration with the German institute.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**I. Technological achievements** 3D-printed personalized Ti-I,plants: • *Precision Manufacturing Workflow:* Development of a high-fidelity integrated workflow: Patient CT data → CAD/CAM virtual surgical planning → Optimized SLS/SLM additive manufacturing → High-precision individualized implant. • *Enhanced Structural Integrity:* Achieving superior mechanical stability and optimized porosity (trabecular structures) to facilitate effective bone ingrowth and long-term implant fixation. Electrochemical surface modification + incorporated functionalized nanodiamonds (NDs) and HAP: • *Programmable Biological Activity*: Successful integration of functionalized NDs and HAP to orchestrate both hard-tissue (bone) and soft-tissue (fibroblast) integration. • *Superior Therapeutic Performance:* Achievement of strong antimicrobial, antioxidant, and osteoinductive

properties through controlled dual-release of bioactive molecules and ions.

- *Clinical Reliability:* Validated biological safety and efficacy through rigorous in-vitro and in-vivo animal model assessments.

Advanced Real-time Biological Monitoring & Analysis:

- *Non-destructive Evaluation:* Implementation of advanced optical technologies, such as Optical Coherence Tomography (OCT) and speckle imaging, to monitor cell proliferation and tissue integration in real-time without damaging the samples.
- *High-Resolution Integration Mapping:* Achievement of precise mapping of the interface between the functionalized titanium surface and regenerating bone/soft tissues through high-resolution histological and bio-imaging analysis.

II. Commercialization potential

Target market and use cases:

- *Maxillofacial & Craniofacial Reconstruction:* Hospitals and specialized surgical centers treating patients with trauma, tumors, or congenital birth defects.
- *Dental & Orthopedic Applications:* Expansion into high-value markets for personalized dental implants and complex orthopedic reconstructions.
- *Aging Population Healthcare:* Addressing the increasing demand for reliable, long-lasting implants for elderly patients with degenerative bone diseases.

Differentiators translate to sales:

- *Reduced Complication Rates:* Strong antimicrobial properties and superior osseointegration leading to fewer revision surgeries and lower post-operative infection risks.
- *Patient-Specific Optimization:* Faster recovery times and improved aesthetic/functional outcomes through anatomical customization, increasing patient and surgeon satisfaction.
- *Cost-Effectiveness:* Streamlined 3D-printing manufacturing reduces lead times and total cost of ownership compared to traditional customized implant fabrication.

Regulatory and market access pathway (joint Korea–Germany advantage):

- Dual-track plan can accelerate global reach:
  Korea: MFDS (KFDA) certification pathway, utilizing local clinical data and domestic reimbursement strategies for rapid market entry.
  Germany/EU: Alignment with EU MDR expectations through Fraunhofer's technical validation, preparing for CE marking and European market distribution.

III. Broader industrial and economic impact

Korea-Germany industrial cooperation:

- Combines complementary strengths:
  Germany: electrochemical surface modification, functionalized nanomaterials with necessary biocompatibile, antimicrobial and osseoinductive properties.
  Korea: world-class high-precision additive manufacturing (SLS/SLM) infrastructure, rapid clinical validation protocols through specialized biomedical research institutes, and a robust medical device ecosystem for agile commercialization.
- Future Platform Template: Establishing a repeatable collaborative model for diverse medical-grade 3D-printing applications beyond craniofacial implants.

Competitiveness of SMEs and high-value manufacturing:

- Upstream Value Creation: Moving from simple assembly to high-margin platform IP and specialized manufacturing of bio-functionalized implants.
- Job Creation & Growth: Strengthening the domestic SME ecosystem in 3D printing, medical software, and advanced biomaterials sectors.

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	Healthcare system and productivity gains: • Economic Savings: Faster rehabilitation and shorter hospital stays translate into significant reduction in long-term public healthcare expenditures and improved patient productivity.

6. References

(1) Rifai, A., Tran, N., Reineck, P., Elbourne, A., Mayes, E., Sarker, A., ... & Fox, K. (2019). Engineering the interface: nanodiamond coating on 3D-printed titanium promotes mammalian cell growth and inhibits Staphylococcus aureus colonization. ACS applied materials & interfaces, 11(27), 24588-24597.

(2) Nemcakova, I., Litvinec, A., Mandys, V., Potocky, S., Plencner, M., Doubkova, M., ... & Sedmera, D. (2022). Coating Ti6Al4V implants with nanocrystalline diamond functionalized with BMP-7 promotes extracellular matrix mineralization in vitro and faster osseointegration in vivo. Scientific Reports, 12(1), 5264.

(3) Gonçalves, J. P., Shaikh, A. Q., Reitzig, M., Kovalenko, D. A., Michael, J., Beutner, R., ... & Opitz, J. (2014). Detonation nanodiamonds biofunctionalization and immobilization to titanium alloy surfaces as first steps towards medical application. Beilstein journal of organic chemistry, 10(1), 2765-2773.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization